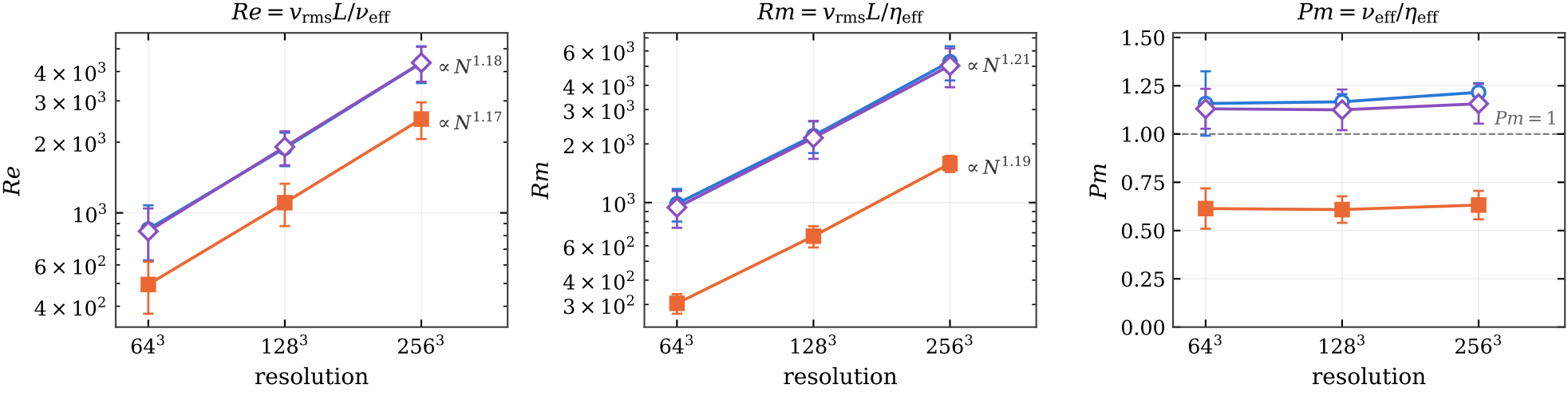
$Re = v_{\mathrm{rms}}L/\nu_{\mathrm{eff}}$
Re
4×10^3
3×10^3
2×10^3
10^3
6×10^2
4×10^2
$\propto N^{1.18}$
$\propto N^{1.17}$
64^3
128^3
256^3
resolution
$Rm = v_{\mathrm{rms}}L/\eta_{\mathrm{eff}}$
Rm
6×10^3
4×10^3
3×10^3
2×10^3
10^3
6×10^2
4×10^2
3×10^2
$\propto N^{1.21}$
$\propto N^{1.19}$
64^3
128^3
256^3
resolution
$Pm = \nu_{\mathrm{eff}}/\eta_{\mathrm{eff}}$
Pm
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$Pm = 1$
64^3
128^3
256^3
resolution
FD: WENO-Z + CT (5th order)
FD: WENO-Z + CT + upwind EMF
FV: VL2 + PLM + GLM (2nd order)
open: $n_K/n_{\mathrm{Nyq}} > 0.8$